

Software Needed:

- PyCharm (not needed but it is what was used here)
- Python
 - Packages: pandas and math
- MySQL
- VirtualBox BIN6970 Linux

Reproduction Instructions:

- 1) With the DBProject Folder there are 5 main folders: Data, Diagrams, Docs, Scripts and SQL
- 2) Clean the raw files in DBProject > Data > Raw with python scripts 1-3. Running it in PyCharm or IDLE.

This will produce the cleaned files:

- data_clinical_patient_cleaned.csv
 - data_mutations_cleaned.csv
 - Data_clinical_sample_cleaned.csv
- 3) Compare the mutation genes with expression genes with script 7
 - a) A report with the comparison will be generated in a txt file
 - 4) Generate the SQL insert files using the python scripts 4-6
These SQL files will be generated:
 - 3_insert_pt_spl.sql
 - 4_mutations_inserts.sql
 - 5_expression_inserts_filtered.sql
 - 5) On your local browser go to: <http://localhost:8888/>
 - a) This should get you on to phpMyAdmin
 - b) Create a new database by clicking New and name it DBProject
 - 6) In the linux command line on virtual box
 - a) Use:
 - i) `mysql -h 127.0.0.1 -P 3306 -u root -pinstructor`
 - ii) use DBProject
 - iii) Source /media/sf_DBProject/(file path for inserts)
Ex: Source /media/sf_DBProject/SQL/1_creatingtables.sql
(1) Do this step for each SQL file in numbered order
 - 7) Wait and verify that the tables have populated correctly
 - a) Check to see the correct number of patients have populated
 - b) Can also check the number of rows in the Genes table, should match numbers seen in the earlier file produced from python script 7